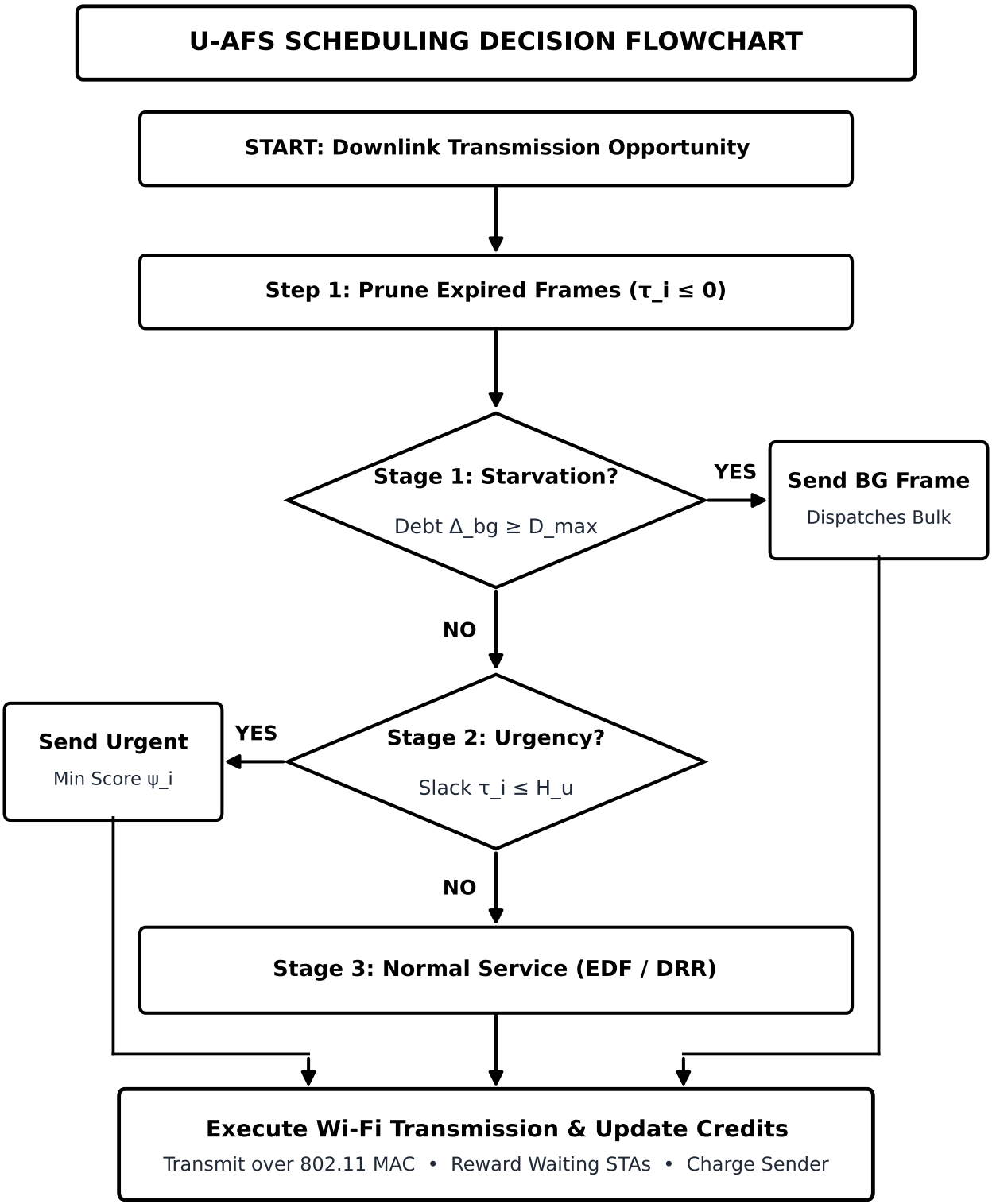
U-AFS SCHEDULING DECISION FLOWCHART
START: Downlink Transmission Opportunity
Step 1: Prune Expired Frames (τ_i ≤ 0)
Stage 1: Starvation?
Debt Δ_bg ≥ D_max
YES
Send BG Frame
Dispatches Bulk
NO
Stage 2: Urgency?
Slack τ_i ≤ H_u
YES
Send Urgent
Min Score ψ_i
NO
Stage 3: Normal Service (EDF / DRR)
Execute Wi-Fi Transmission & Update Credits
Transmit over 802.11 MAC • Reward Waiting STAs • Charge Sender